

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus II

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Evaluate: $\int_0^{\pi} \int_x^{\pi} \frac{\sin y}{y} dy dx$ by changing the order of integration. 5
 - b) Evaluate the integrals: $\iiint_V (x^2 + y^2 + z^2) dx dy dz$ taken over the volume of the sphere $x^2 + y^2 + z^2 = 1$. 5
 - c) Find the volume of the solid whose base is the region in the xy-plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$. 5
 2. a) Solve the differential equation: $y'' + 9y = 0$ by using power series method. 7
 - b) Define Legendre differential equation. Also, derive the solution of Legendre differential equation. 8
- OR**
- If $J_n(x)$ represents the Bessel's function of order n , then show that:
- i. $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$ 4
 - ii. $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ 4
3. a) Solve the following differential equation by using Laplace transform $y'' + 2y' - 3y = 6e^{-2t}$, $y(0) = 2$, $y'(0) = -14$. 7
 - b) i) State second shifting property for Laplace transform and using this, find the Laplace transform of $e^{-3t} u(t-3)$. 4
 - ii) Find the inverse Laplace transform of $\frac{\pi}{(s-3)^2 + \pi^2}$. 4
 4. a) Find the directional derivative of $f = x^2 yz + 4xz^2$ at $(1, -2, 1)$ in the direction $2\vec{i} - \vec{j} - 2\vec{k}$. 5

- b) Show that the vector $\vec{F} = (x^2 - yz, y^2 - xz, z^2 - xy)$ is conservative vector field and find the scalar potential function ϕ such that $\vec{F} = \nabla \phi$. 5
 - c) Find the work done by the force $\vec{F} = (3x^2, 2xz - y, z)$ along the curve $x = t$, $y = \frac{t^2}{4}$, $z = \frac{3t^2}{8}$ from $(0, 0, 0)$ to $(2, 1, 3)$. 5
 5. a) Evaluate the integral by using Green's theorem 7
- $\oint_C [(3x^2 + y) dx + 4y^2 dy]$ where C : the boundary of the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 2)$; oriented in counterclockwise direction.
- b) State Stoke's Theorem. Evaluate $\oint_C (\vec{F} \cdot d\vec{r})$ by using Stoke's theorem, where $\vec{F} = (y, xz^3, -zy^3)$ and $C: x^2 + y^2 = 4, z = 3$. 8
- OR**
- Evaluate: $\iint_S \vec{F} \cdot \vec{n} dA$, where $\vec{F} = (18z, -12, 3y)$ and S is the surface of plane $2x + 3y + 6z = 12$ in the first octant.
6. a) Find the Fourier series of $f(x) = x + |x|$, $-\pi < x < \pi$ and show that $\frac{\pi^2}{8} = 1 + \frac{1}{9} + \frac{1}{25} + \dots$ 7
 - b) Find the Fourier sine and cosine series for the function $f(x) = x^2$ in the interval $0 < x < L$. 8
 7. Write short notes on: (Any two) 2×5
- a) Find the general solution of the partial differential equation $u_x + u_y = u$.
 - b) Derive the one-dimensional traffic flow model using conservation law.
 - c) A particle moves along the curve $x = t^3 + 1$, $y = t^2$, $z = 2t + 5$. Find the velocity and acceleration at $t = 1$.